GDAs-OT: A Prediction Method of Gene-Disease Associations Based on Optimal Transport for Identifying Genes Related to Immune-Related Adverse Events Supplementary Material

Ruhao Liu[1] , **Suixue Wang**[1] , **Hang Yu**[2] , **Peng Li**[3] * and **Qingchen Zhang**[4] *
[1]School of Information and Communication Engineering, Hainan University, Haikou, China
[2]School of Engineering, University of Warwick, Coventry, CV4 7AL, UK
[3]School of Computer Science and Technology, Dalian University of Technology Key Laboratory of Social Computing and Cognitive Intelligence, Dalian University of Technology
[4]School of Computer Science and Technology, Hainan University, Haikou, China
{lrh, wangsuixue, zhangqingchen}@hainanu.edu.cn, Hang.Yu.1@warwick.ac.uk, pli@dlut.edu.cn

1 Biomedical Pre-trained Language Models Selection

To obtain high-quality gene and disease representations, we investigate the effectiveness of three biomedical pre-trained language models-BioGPT [Luo *et al.*, 2022], LinkBERT [Yasunaga *et al.*, 2022], and PubMedBERT [Gu *et al.*, 2021]-for node feature construction. Specifically, each model is used to encode gene- and disease-related biomedical texts into vector representations, which serve as the initial node features in the gene-disease graph. These representations are subsequently fed into the GDAs-OT framework for experimental evaluation under identical experimental settings. The corresponding results are presented in Figure 1.

As shown in Figure 1, the node features generated by PubMedBERT consistently achieve superior performance compared with those produced by BioGPT and LinkBERT across all evaluation metrics. This observation suggests that PubMedBERT is able to capture more informative semantic representations relevant to gene-disease associations. Consequently, considering both predictive performance and representation quality, PubMedBERT is selected as the final feature encoder in our framework.

2 Squared Loss vs Kullback-Leibler Divergence

In the main text, two different loss functions are adopted for solving the Gromov-Wasserstein (GW) discrepancy [Peyré *et al.*, 2016], namely the squared loss and the Kullback-Leibler (KL) divergence. Their formulations are given by

$$L(a,b) = \frac{1}{2}|a-b|^2 \tag{1}$$

and

$$L(a,b) = a\log\frac{a}{b} - a + b, \tag{2}$$

respectively. When computing

$$C_\kappa = f_1(C^g)p_g 1_\xi^\top + 1_\eta (p_d)^\top f_2(C^d)^\top, \tag{3}$$

the corresponding functions are specified as follows. For the squared loss, we have $f_1(k) = k^2$, $f_2(l) = l^2$, $h_1(k) = k$ and $h_2(l) = 2l$. For the KL divergence, the functions are defined as $f_1(k) = k\log(k) - k$, $f_2(l) = l$, $h_1(k) = k$ and $h_2(l) = \log(l)$. We evaluate the performance of GDAs-OT under these two loss functions, and the results are reported in Figure 2. Based on the experimental results, the squared loss is adopted in this study.

*Corresponding author

References

[Gu *et al.*, 2021] Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare (HEALTH)*, 3(1):1–23, 2021.

[Luo *et al.*, 2022] Renqian Luo, Liai Sun, Yingce Xia, Tao Qin, Sheng Zhang, Hoifung Poon, and Tie-Yan Liu. Biogpt: generative pre-trained transformer for biomedical text generation and mining. *Briefings in Bioinformatics*, 23(6):bbac409, 2022.

[Peyré *et al.*, 2016] Gabriel Peyré, Marco Cuturi, and Justin Solomon. Gromov-wasserstein averaging of kernel and distance matrices. In *International Conference on Machine Learning*, pages 2664–2672. PMLR, 2016.

[Yasunaga *et al.*, 2022] Michihiro Yasunaga, Jure Leskovec, and Percy Liang. Linkbert: Pretraining language models with document links. *ArXiv Preprint ArXiv:2203.15827*, 2022.

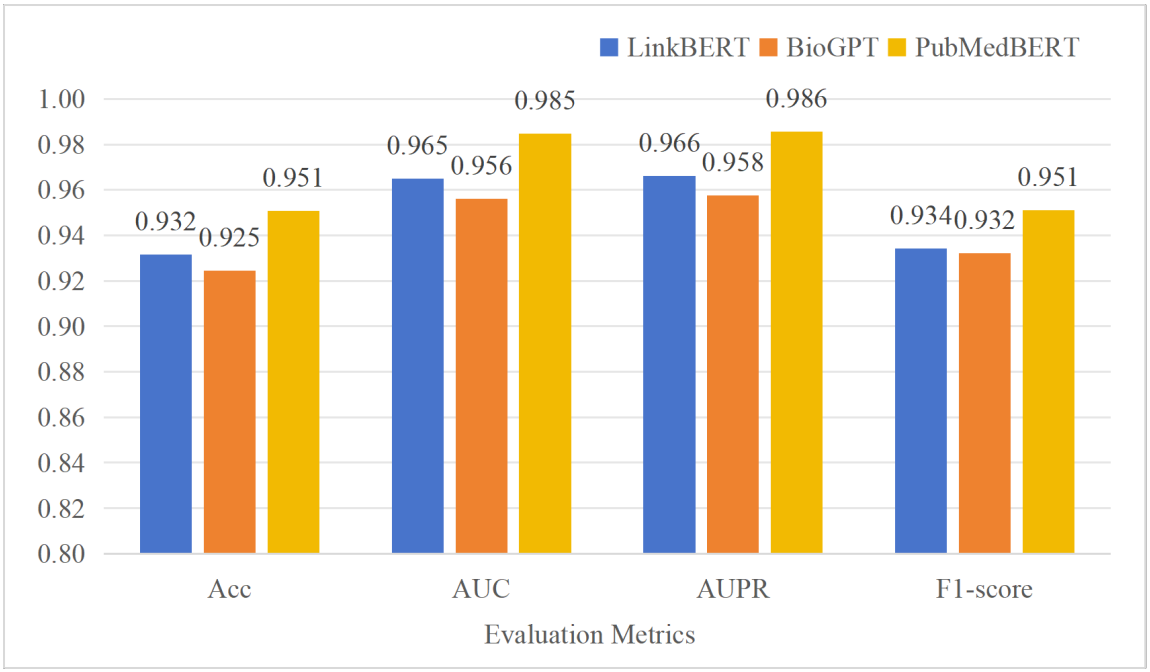

Figure 1: Comparison of different biomedical pre-trained language models.

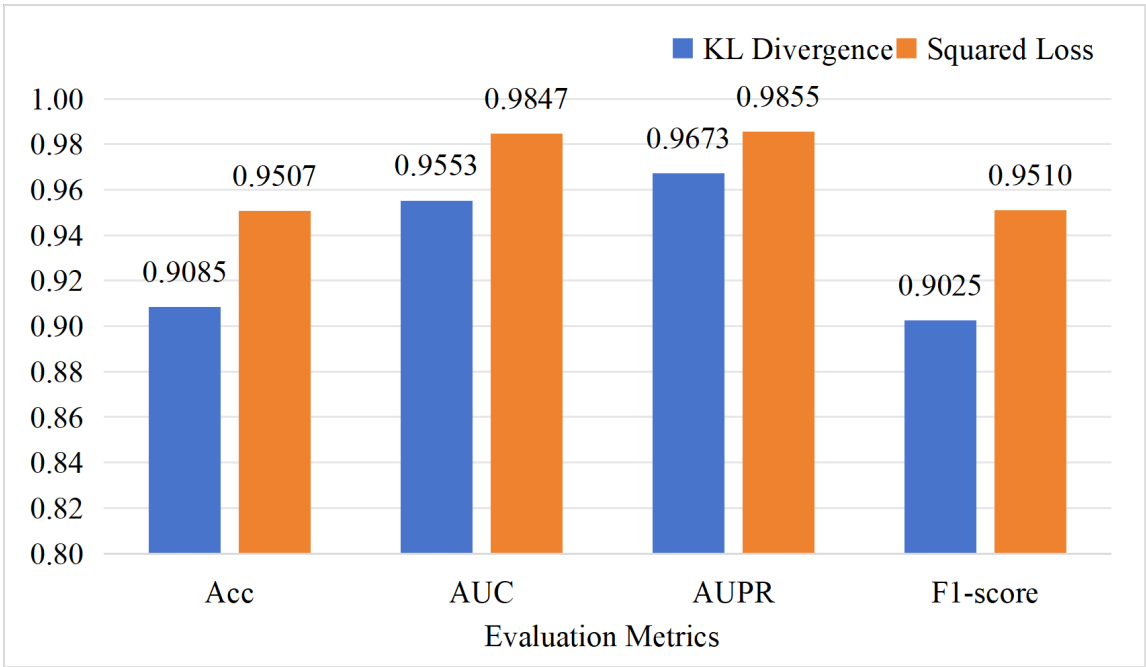

Figure 2: Comparison between KL divergence and squared loss.

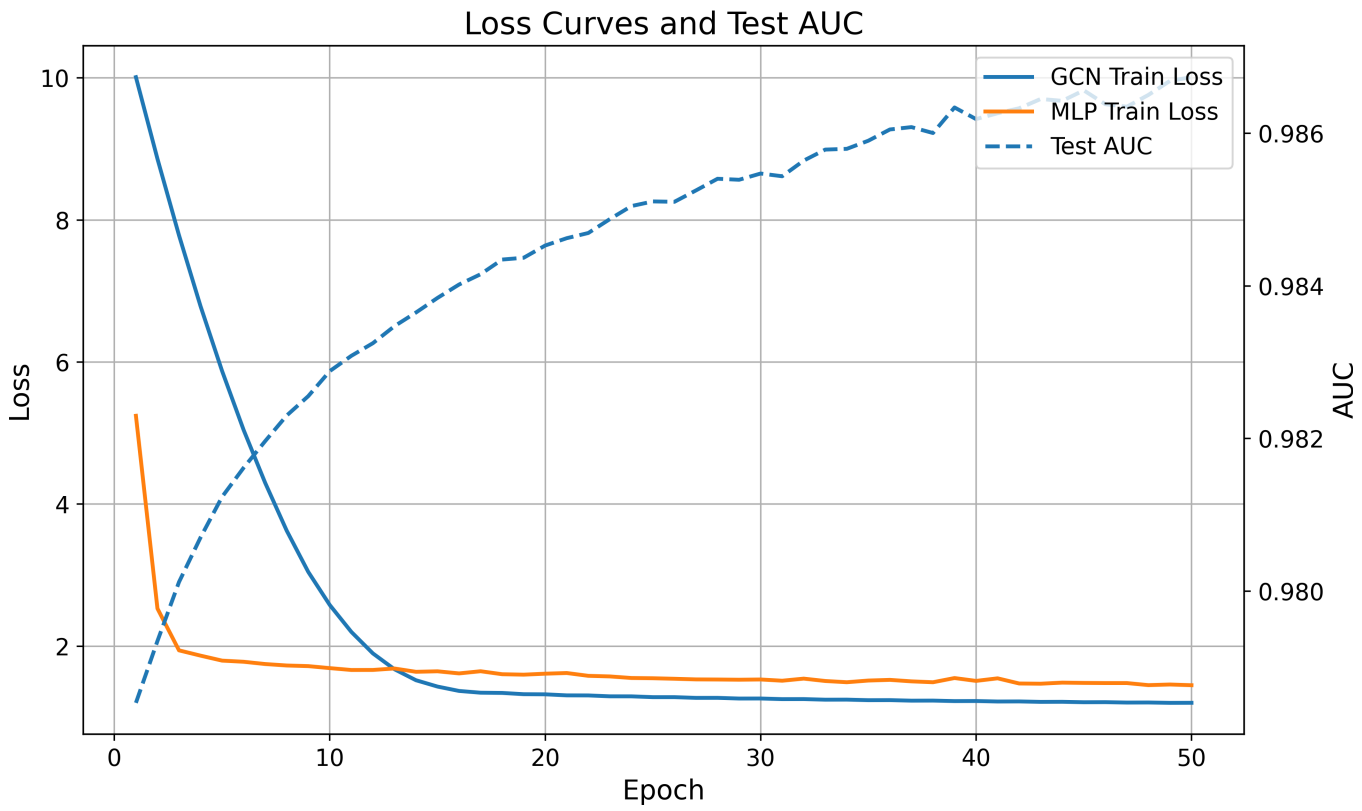

Figure 3: Loss curve and test AUC

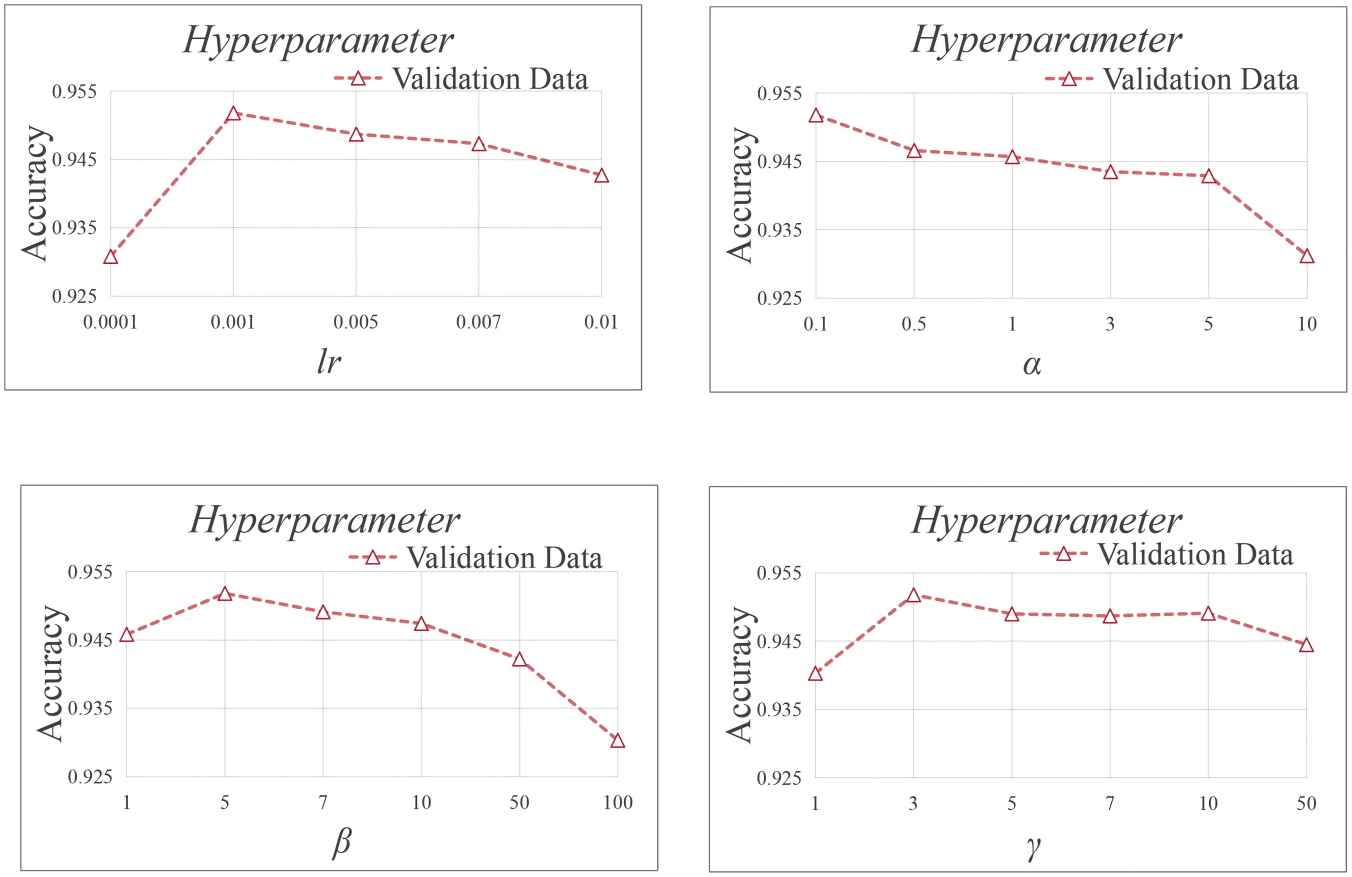

Figure 4: Parameter sensitivity analysis.